



3.24pt

Introduction

Many physical systems are governed by dynamics that permit more than one state, such as the glacial/interglacial cycle of our climate. Moreover, a bifurcation in the dynamics may change the number of permitted states.

A dramatic and relatively sudden shift from one state to another is known as a tipping point. Hopefully the time series from a measurement of the system will reveal the tipping point.

Project Aims

There are two possible goals whose paths of approach are intertwined:

1. Detecting a tipping point
2. Predicting a future tipping point

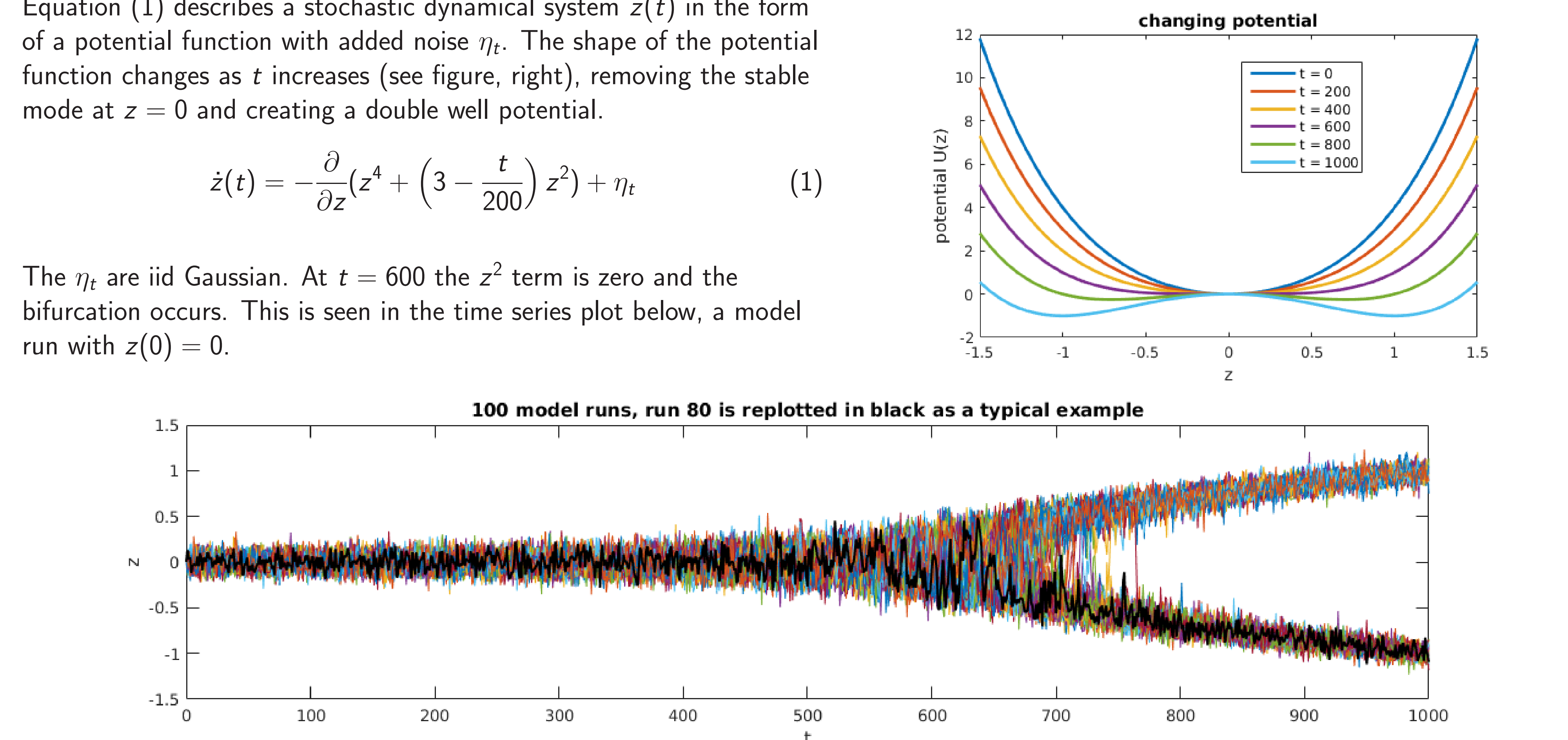
This PhD project aims to develop a multi-purpose prediction tool for tipping points in physical systems. Several techniques are used to analyse time series for Early Warning Signals.

A Simple Example of a Tipping Point

Equation (1) describes a stochastic dynamical system $z(t)$ in the form of a potential function with added noise η_t . The shape of the potential function changes as t increases (see figure, right), removing the stable mode at $z = 0$ and creating a double well potential.

$$\dot{z}(t) = -\frac{\partial}{\partial z}\left(z^4 + \left(3 - \frac{t}{200}\right)z^2\right) + \eta_t \quad (1)$$

The η_t are iid Gaussian. At $t = 600$ the z^2 term is zero and the bifurcation occurs. This is seen in the time series plot below, a model run with $z(0) = 0$.

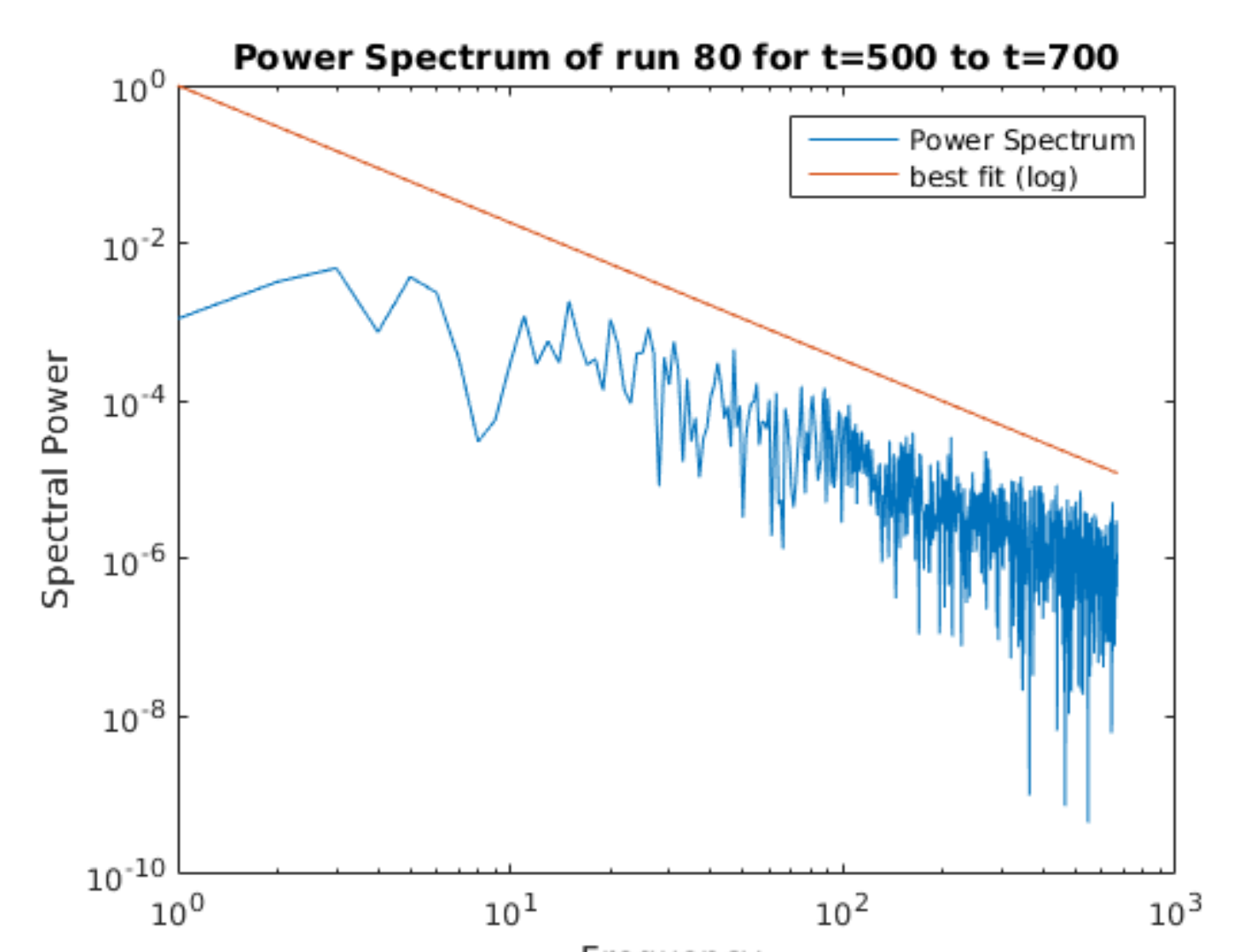


Measuring Autocorrelation in a Time Series

It is proposed that a a tipping point visible in a time series will be anticipated by a rise in the series' autocorrelation [1,2], as a forcing pushes the system towards the tipping point. Equation (2) gives the Auto-Correlation Function with a lag of 1 (ACF₍₁₎), for a time series x .

$$ACF_{(1)}([x_i]_1^N) = \frac{\frac{1}{N-1} \sum_{i=2}^N (x_{i-1} - \bar{x})(x_i - \bar{x})}{\text{Var}(x)} \quad (2)$$

Detrended Fluctuation Analysis (DFA) also provides an indicator of autocorrelation [3]. In the continuous, stationary case, the DFA coefficient is linearly related to the ACF.



We may also find the Scaling Exponent of the series' power spectrum (or PSSE). In the figure (left) the power spectrum is plotted for the part of the series (using the Equation (1) example) where the bifurcation occurs. The slope of the line is -1.74 , giving a PSSE of 1.74, similar to Brownian Motion. For the first part of the series the PSSE is ≈ 0 , which is the same as for white noise.

The PSSE is also (in the stationary, continuous case) linearly related to the ACF and DFA coefficients.

Tipping Point Prediction: Fingerprinting

To detect a rise in autocorrelation, which may give an Early Warning Signal for a tipping point [1], we calculate possible indicators in a sliding window of $\sim N/5$ points.

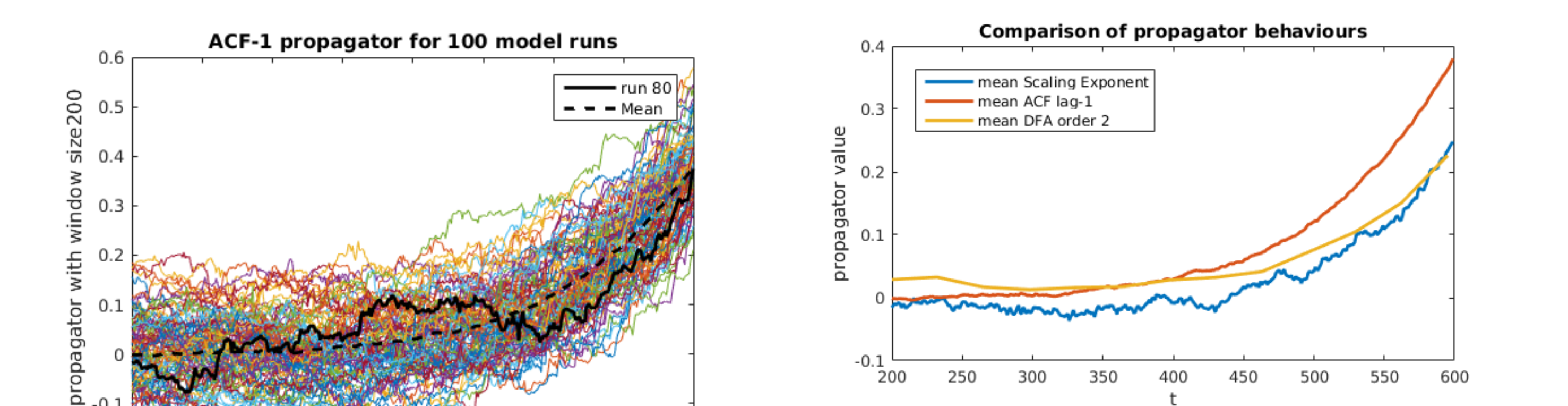
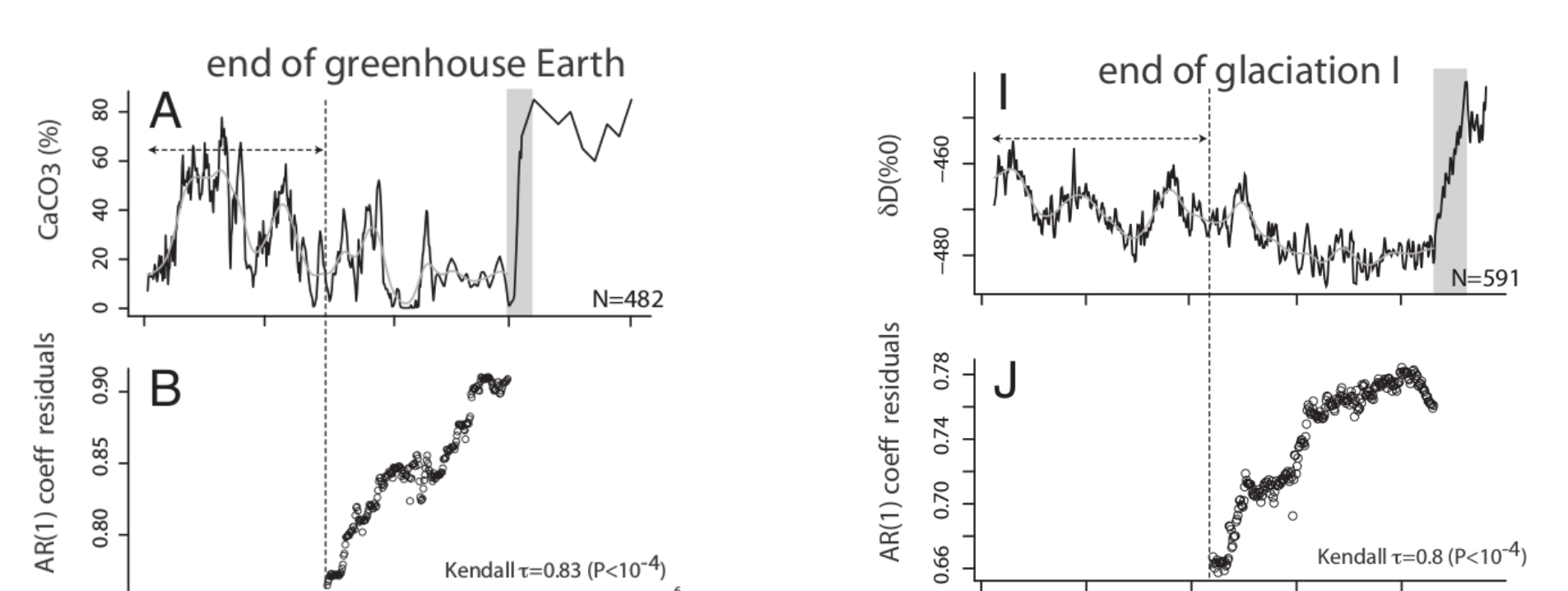


Figure: The ACF lag-1 is calculated (see Equation (2)) for 100 model runs upto the point of bifurcation ($t = 600$).

In Figure 2 we see that all indicators begin to rise at $t \approx 400$, at $t = 500$ an upward trend is clear. In this case, all three provide a good EWS. In many situations we may not be able to take the mean of 100 experiments, certainly for predicting a geophysical event we will have only one time series. In Figure 1 we see that the ACF signal is not so clear in this case, it is thus worth considering all indicators.

Applications of Fingerprinting

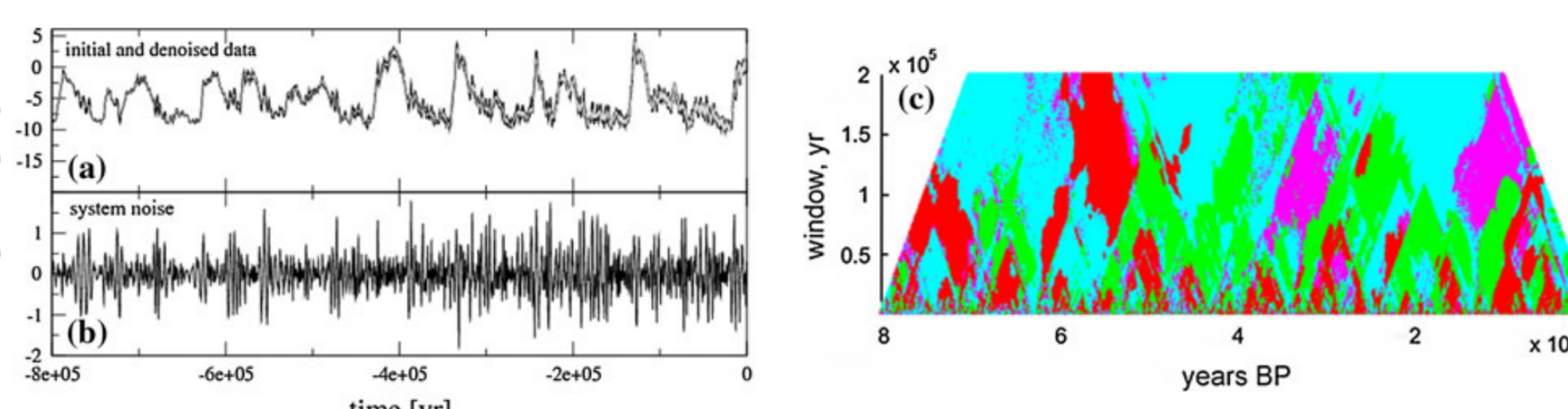
The fingerprinting technique is applied to several time series by Dakos et al. [4], in this paper yet another indicator for auto-correlation is used: an autoregressive model of the form $x_{t+1} = \alpha x_t + \eta_t$ is fitted to the data using ordinary least squares, and the coefficient α is used as an indicator.



These figures (taken from [4]) show two examples, in both cases the indicator shows a significant trend sometime before the tipping point. Many more examples of tipping points can be found in [7].

Potential Analysis: Detection of Tipping Points

One aim of tipping point research is detection. One approach (as in [5]) is to assume that a system resembles a n -well potential system, with a form similar to Equation (1), and to fit such an equation to a window of the time series.



The figures (taken from [5]) show Antarctic ice-core temperature proxy data (left) and a contour plot showing the estimated number of potential wells of the best-fitting model: red - one; green - two; cyan - three; purple - four wells. The shape of the fitted potential function depends on the window size (y -axis of contour). This example does not show a clear bifurcation (as in the Equation (1) example) but a patchy pattern. Livina et al. [5] show that further knowledge of a physical system may be necessary.

Tipping Points in Multi-dimensional Time Series

It is desirable to detect and predict tipping points in more than one dimension. This can relate to

1. Multiple variables: more than one variable (e.g. temperature and pressure) may be relevant.
2. A variable measured at multiple points in space: a tipping point may affect a large area, or an unknown area within a given region.

In both cases, providing more data may give a more reliable EWS.

In [1], EOF is used to reduce the dimension of a system. It is also possible to consider several dimensions at once, in [6], a two-dimensional analogue of Equation (2) is formulated.

It is likely that knowledge of the physical system will be needed to determine which set of variables may be relevant.

References

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